

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			more acid added than is needed to react with all the marble chips accept marble chips run out <u>first</u> / acid is in excess		1		1		1
	(b)			line drawn steeper / to the left of line A <u>and</u> reaching a maximum level of 60 cm ³		1		1		
	(c)			8.66 (2) if answer incorrect award (1) for either of following $\frac{7.8}{100} / 0.078 / 7.8 \times 111$ $0.078 \times 111 / \frac{866}{100}$ ecf possible alternative method $n(\text{CaCO}_3) = 0.078 \text{ mol} \Rightarrow n(\text{CaCl}_2) = 0.078 \text{ mol} \quad (1)$ $\text{mass CaCl}_2 = 0.078 \times 111 = 8.66 \text{ g} \quad (1)$ ecf possible		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			26.6 (2) if answer incorrect award (1) for either of following $\frac{14.3}{53.7} / 0.266$ 0.266×100 no ecf possible alternative methods $\frac{14.3}{0.537}$ (1) $\frac{100}{53.7} \times 14.3$ (1) 26.6 (1) 26.6 (1) no ecf possible		2		2	2	
	(e)			a catalyst lowers the minimum energy needed for reaction / successful collisions / lowers the activation energy (1) this means more <u>successful</u> collisions (per second) / higher frequency/chance of <u>successful</u> collisions (so higher rate) (1) if no reference to collisions award (1) mark for 'a catalyst provides a suitable surface for a reaction'	2			2		
				Question 5 total	2	6	0	8	4	1